Sustainability of the Pulseq environment and community building

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Vendor-Agnostic Pulse Sequence Programming with Pulseq II

ISMRM Virtual Meeting

Feb 25, 2026

Agenda

1. Pulseq: The journey so far

2. Challenges & Futures
   a. Toolkit coordination (MATLAB, Python, etc.)
   b. Community building and central structures
   c. Regulatory certification (FDA/CE) & enabling third-party commercialization
   d. Sustainability (financial & people power)

Goal: get your feedback and ideas to help prioritize and plan next steps

1. Pulseq: The journey so far

High visibility in the ISMRM community

Established prototyping platform

Over 200 Siemens sites

Over 100 GE users

United Imaging, Philips, MR Solutions, Home-built MR scanners…

⇒ *How to keep the momentum going forward?*

2.a Toolkit coordination (MATLAB, Python, …)

Challenges

- Possible fragmentation across toolkits; duplicated effort.
- Differences in documentation, support, and functionality between toolkits.

Where we are now

- MATLAB and Python developers “keeping an eye on each other”. Sustainable?

Opportunities & Futures

- Create cross-platform documentation and tutorials/examples.
- How do we coordinate these efforts? Is it needed?
- … Ideas? Please post in Q&A!

Poll 1

Should we (the community) set up a repository with cross-platform tutorials/examples? *Yes / No pressing need*

Should we (the community) establish (and maintain) a cross-platform toolkit API specification? *Yes / No pressing need*

Would you be willing to volunteer to help organize such efforts? *Yes / Not now*

[Result will be posted on the Github site]

2.b Community building and central structures

Challenges

- Do we need a central organization? Formalized or just as is?
- Determining where central organization should reside (and how centralized it should be).
- Scattering of resources; lack of clarity about where to start learning about Pulseq and how to best engage with the community.
- Define clear pathways to become involved.

2.b Community building and central structures

Where we are now:

- User discussion and bug report forums, https://github.com/pulseq/pulseq
- Tutorials and prior workshop/webinar materials, https://github.com/pulseq
- Central registry for community-contributed projects, https://harmonizedmri.github.io ✅
- Monthly Pulseq user group meeting, https://harmonizedmri.github.io/
- Mailing list is set up -- should we use it? https://groups.google.com/g/harmonizedmri

- Many other scattered resources
- Somewhat unclear where to start learning about Pulseq
- Lack of clear pathways to become involved ❌

2.b Community building and central structures

Opportunities & Futures

- Establish a dedicated ISMRM study group for Pulseq
  - Alternative 1: Collaborate with/within the existing OpenScience study group
  - Alternative 2: Establish an organisation outside ISMRM
- Host regular hackathons
- Offer regular teaching sessions and workshops
- Develop new/more structured recognition pathways
- Solicit community feedback (surveys etc)
- …

[Poll 2]

2.c Regulatory certification (FDA/CE) & enabling third-party commercialization

Where we are now

- Open-source licensing supports broad adoption (including commercial)
  - Thanks to all contributing to the PyPulseq license switch!
- No FDA/CE certification; not cleared for clinical use
- Some collaborative projects, but limited integration into clinical workflows
- Documentation aimed at research, not regulatory compliance
- No established pipeline for clinical approval
- No structures for support / post-market surveillance

2.c Regulatory certification (FDA/CE) & enabling third-party commercialization

Opportunities & Futures

- Establish roadmap for FDA/CE certification
- Collaborate with clinical and industry partners for pilot studies
- Develop regulatory-ready documentation and testing
- Enable certified, commercial third-party Pulseq applications across vendors
- Create IP guidelines supporting both open-source and proprietary innovation
- Establish an organization to facilitate the above processes?

2.c Regulatory certification (FDA/CE) & enabling third-party commercialization

Do we need an organisation (like Mozilla Foundation)?

Recent report of a company moving away from Pulseq due to a missing pathway towards medical device approval

Do we need an organization to pursue FDA-approval or CE-Certification?

What is the vendor perspective

We know Siemens will provide Open Sequence, so Pulseq could dock on it as a certified interface

GE? United Imaging? Others?

Poll 3

Do you think there is a need for a Pulseq organization to pursue FDA/CE certification? *Yes / Not yet*

Would you be willing to volunteer for such an organization? *Yes / Not yet*

[Result will be posted on the Github site]

2.d Sustainability (financial & people power)

Challenges

- Reliance on volunteers risks burnout and limits to long-term momentum
- Difficulty securing consistent funding
- Limited incentives for contributors (academics, industry, students) to invest their time
- Uncertainty about resource allocation for infrastructure, documentation, and community events

2.d Sustainability (financial & people power)

Where we are now

- Matlab-Pulseq: small volunteer team (mainly Freiburg and Michigan groups)
  - NIH, EU and EURAMET funding,
    mostly running out….

- PyPulseq: large volunteer team
  - NIH R03; otherwise mostly unfunded

- Contributor recognition
  - http://harmonizedmri.github.io
  - Monthly Pulseq user group meeting/seminar series

U24 project

A harmonized vendor-agnostic environment for multi-site **functional MRI** studies

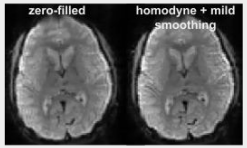
NIH U24NS120056

Freely available protocols

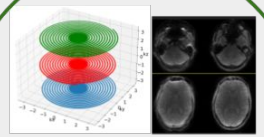
Paid support is available

Centralized IRB (single IRB)

https://github.com/HarmonizedMRI/Functional



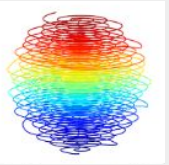
Single/multi-echo SMS-EPI; 3D/4D EPI
https://github.com/HarmonizedMRI/Functional



3D stack-of-spirals
https://github.com/HarmonizedMRI/SOSP3d

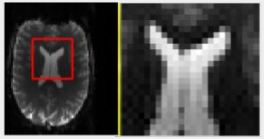
python

MR encephalography (MREG; ultrafast fMRI)

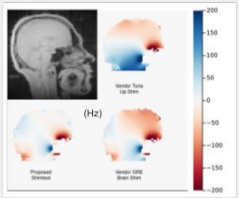


Pulseq

Pulseq platform (standard, toolbox, interpreters)
https://pulseq.github.io/



Quality Assurance
https://github.com/HarmonizedMRI/qualityAssurance



B0 shimming
https://github.com/HarmonizedMRI/B0shimming

source

2.d Sustainability (financial & people power)

Opportunities & Futures

- Pursue grant funding and industry partnerships
  - Consider including Pulseq tool developers as collaborators
- Offer paid support, consulting, or training
- Implement contributor recognition (credit, awards, leadership)
- Establish transparent governance for resource allocation
- …

Poll 4

Would you/your team consider budgeting for paid support for a Pulseq-based protocol in your next grant proposal? *Yes / No need*

If yes, what would you consider a reasonable maximum annual fee? *$1k / $5k / $10k / $15k*

[Result will be posted on Github]

What’s next

“Easy” first steps you can do:

1. Join the monthly user group meeting, https://harmonizedmri.github.io/
2. Post questions in the discussion board, https://github.com/pulseq/pulseq/discussions
3. Submit bug reports, https://github.com/pulseq/pulseq/issues
4. Volunteer to answer questions
5. Contribute your projects to https://harmonizedmri.github.io/

Coordinated action:

- Let’s discuss at the next user group meeting (Mar 6)

Sustainability beyond U24 funding period (M3.5)

Proposal:

- Paid annual subscription + optional service. Managed by UM Innovation Partnerships.
- For use of our fMRI, B0 shimming, and QA/QC protocols in funded research studies
- Request that users budget for this in grant proposal (we provide template text)

Free to use (MIT license)

- Develop new/derived methods
- Evaluate existing methods
- Collect pilot data

Paid annual subscription (service package)

- Funded research studies where primary aim is not method development/evaluation
- Tiered support

INNOVATION PARTNERSHIPS